LESSON 6.7

Find Factors and Multiples

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the greatest common factor and the least common multiple of two numbers, and use each to solve a real problem.

WHY THIS MATTERS The station quartermaster has two jobs today. First, split the pencil and notebook stock into identical combo packs with nothing left over — that is a factor question. Second, work out when two supply runs that repeat on different cycles will land in the same week — that is a multiple question. Same lesson, opposite tools. Today the stock is 24 pencils and 36 notebooks. One supply run arrives every 6 days and another arrives every 8 days.

REMEMBER

Find Factors and Multiples

Greatest Common Factor (GCF) & Least Common Multiple (LCM) A factor divides a number evenly, so factors answer questions about splitting a stock into equal groups. A multiple is where you land when you skip-count, so multiples answer questions about when repeating events line up.

1. GCF: Largest factor shared by two numbers — used to simplify fractions or divide into equal groups.
2. LCM: Smallest shared multiple — used to find common denominators.
3. Use prime factor trees or list method to find common factors/multiples

Words: Factor *(Factor)* · Multiple *(Múltiplo)* · Greatest common factor (GCF) *(Máximo común divisor (MCD))* · Least common multiple (LCM) *(Mínimo común múltiplo (mcm))* · Factor pair *(Par de factores)*

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which expression is equivalent to 5x + 2x?

(A) 7x
(B) $7x^2$
(C) 10x
(D) 52x

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which expression is NOT equivalent to 2y + 2y + 2y?

(A) $2y^3$
(B) 2y × 3
(C) 3(2y)
(D) 6y

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Are 3(n + 2) and 3n + 6 equivalent? Test with n = 4.

(A) No — one is 18 and one is 14
(B) Yes — both equal 18
(C) Yes, but only when n = 0
(D) You cannot test with a number

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Simplify 6a + 4 − 2a.

(A) 2a + 4
(B) 4a + 4
(C) 4a
(D) 8a

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The school store has 48 pencils and 36 notebooks. Every combo pack must hold the same number of pencils and the same number of notebooks, using all the stock with nothing left over. What is the GCF of 48 and 36?

(A) 4
(B) 6
(C) 12
(D) 144

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

The school store has 48 pencils and 36 notebooks. Every combo pack must hold the same number of pencils and the same number of notebooks, using all the stock with nothing left over. What is the LCM of 6 and 8?

(A) 2
(B) 14
(C) 24
(D) 48

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Both problems say "common." How do you decide whether a problem wants the GCF or the LCM?

SENTENCE STARTER If the problem ___ something into equal groups, I use the ___. If it asks when things ___ again, I use the ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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WHY THIS MATTERS The station quartermaster has two jobs today. First, split the pencil and notebook stock into identical combo packs with nothing left over — that is a factor question. Second, work out when two supply runs that repeat on different cycles will land in the same week — that is a multiple question. Same lesson, opposite tools. Today the stock is 24 pencils and 36 notebooks. One supply run arrives every 6 days and another arrives every 8 days.

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each pair to its greatest common factor.

_____ 12 and 18	(A) 6
_____ 48 and 36	(B) 12
_____ 10 and 15	(C) 5
_____ 7 and 9	(D) 1

2 **Write the name of the correct group on each line.**

Sort each statement about factors and multiples.

Groups: True of factors | True of multiples

_____ Never larger than the number

_____ Never smaller than the number

_____ There is a limited number of them

_____ The list goes on forever

_____ Found by dividing

_____ Found by skip-counting

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

The school store has 48 pencils and 36 notebooks. Every combo pack must hold the same number of pencils and the same number of notebooks, using all the stock with nothing left over. A problem asks when two repeating schedules meet again. Which do you use?

(A) The least common multiple

(B) The greatest common factor

(C) Either one works

(D) Neither — you list factors

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Is 4 a factor or a multiple of 12?

(A) A factor

(B) A multiple

(C) Both

(D) Neither

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is the GCF of 10 and 15?

(A) 1

(B) 5

(C) 10

(D) 30

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Find the error

1. **Problem** List the factors of 16.
2. **Step 1** 16 = 1 × 16, 2 × 8, 4 × 4
3. **Step 2** So the factors are 2, 4, and 8.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Both problems say "common." How do you decide whether a problem wants the GCF or the LCM?

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How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help